

PELICAN

ENGINEERING PORTFOLIO

OUR

MARTON PETER KISS

Suspension, chassis and powertrain design for Formula Student. Four seasons, three car generations, 100+ mechanical parts taken from concept to a race-proven car.



Capability

Tools, certification and education

Mechanical design engineer working on suspension, chassis and powertrain hardware for Óbuda University Racing Team. Every part shown here was modelled in SolidWorks, validated in ANSYS, detailed to a manufacturing drawing with fits and tolerances, and run on the race car.

DESIGN AND ANALYSIS TOOLS

SolidWorks	Professional, certified
3DExperience	Working
ANSYS Mechanical	Working
Lotus Suspension Analysis	Working
Lotus Engine Simulation	Working
MAXXECU MTune, MoTeC M1	ECU calibration and telemetry
MATLAB / Simulink	Onramp
Python	Intermediate

CERTIFICATION AND PUBLISHED WORK

- SolidWorks Professional, CSWP
- The Complete Guide to Vehicle Dynamics in F1
Racetrack Dynamics, 2025 — course led by F1 engineers
- Siemens Plant Simulation, certified
- Scientific Students' Associations, third place
Powertrain design for a Formula Student Race Car
- Rear Upright Developed for a Formula Student Racing Car
Engineering Symposium at Bánki, ESB 2023
- Design of an Exhaust Primary Manifold for a Formula Student Race Car
31st National Mechanical Engineering Conference
- Rear Wheelhub Design of a Formula Student Racecar
32nd National Mechanical Engineering Conference
- Optimisation of the Pipe Chassis for a Formula Student Race Car
InnoVetas, Gép Journal, 2024

EDUCATION

Óbuda University

MSc Machine Design, from 2026 · BSc Vehicle Engineering, 2021-2025

Thesis

Development and design of a Formula Student suspension system

Selected work

Three subsystems and the build

- 01

Suspension

REAR CORNER ASSEMBLY | OUR003 | 2025
- 02

Chassis

TUBULAR SPACEFRAME | OUR003 | 2024-2025
- 03

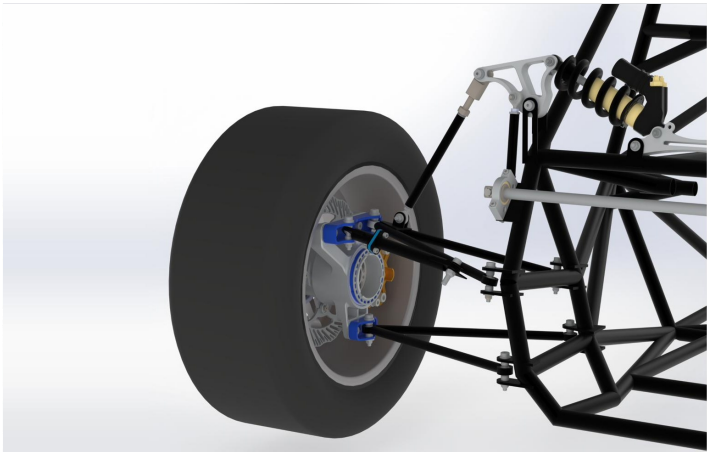
Powertrain

INTAKE, EXHAUST, DRIVETRAIN | OUR003 | 2025
- 04

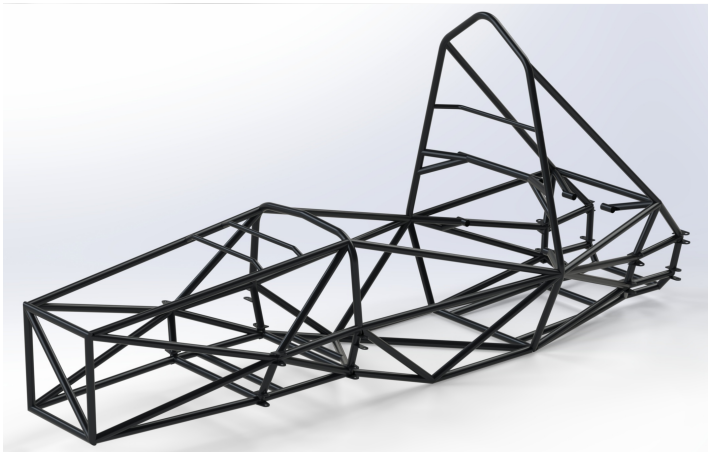
Manufacturing

DESIGN FOR MANUFACTURE, CNC TURNING, MILLING

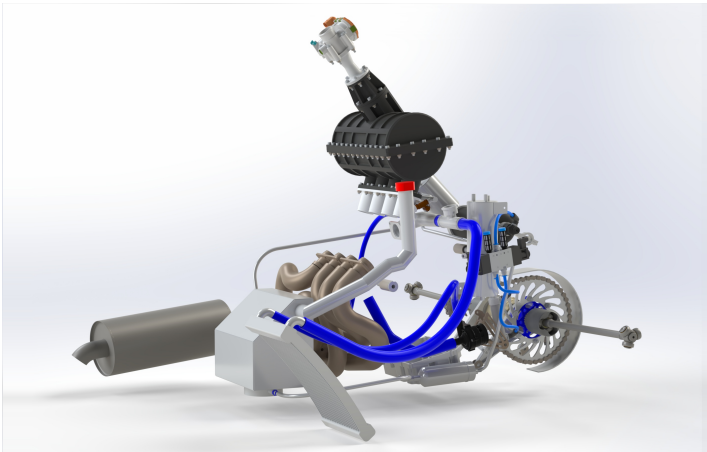
100+ MECHANICAL PARTS
OUR001 - OUR003 | 2021-2026



01 Suspension



02 Chassis



03 Powertrain



04 Manufacturing

Rear suspension

Óbuda University Racing Team · OUR003 · 2025

The upright connects to two brackets giving two-way camber adjustability. The upper bracket carries both the upper A-arm and the tie rod, tying the tie rod into the A-arm for optimised stress flow, better adjustability and simpler assembly. The damping system has a 1.4 motion ratio making mechanical connection with a machined aluminium rocker which is also connected to the anti-roll bar. The chassis nodes are the hardpoints taken from a kinematic model in which the roll centre has been determined with aerodynamic loads kept in mind. The rear corner assembly runs 10% less unsprung mass than the previous generation, with no structural compromise.

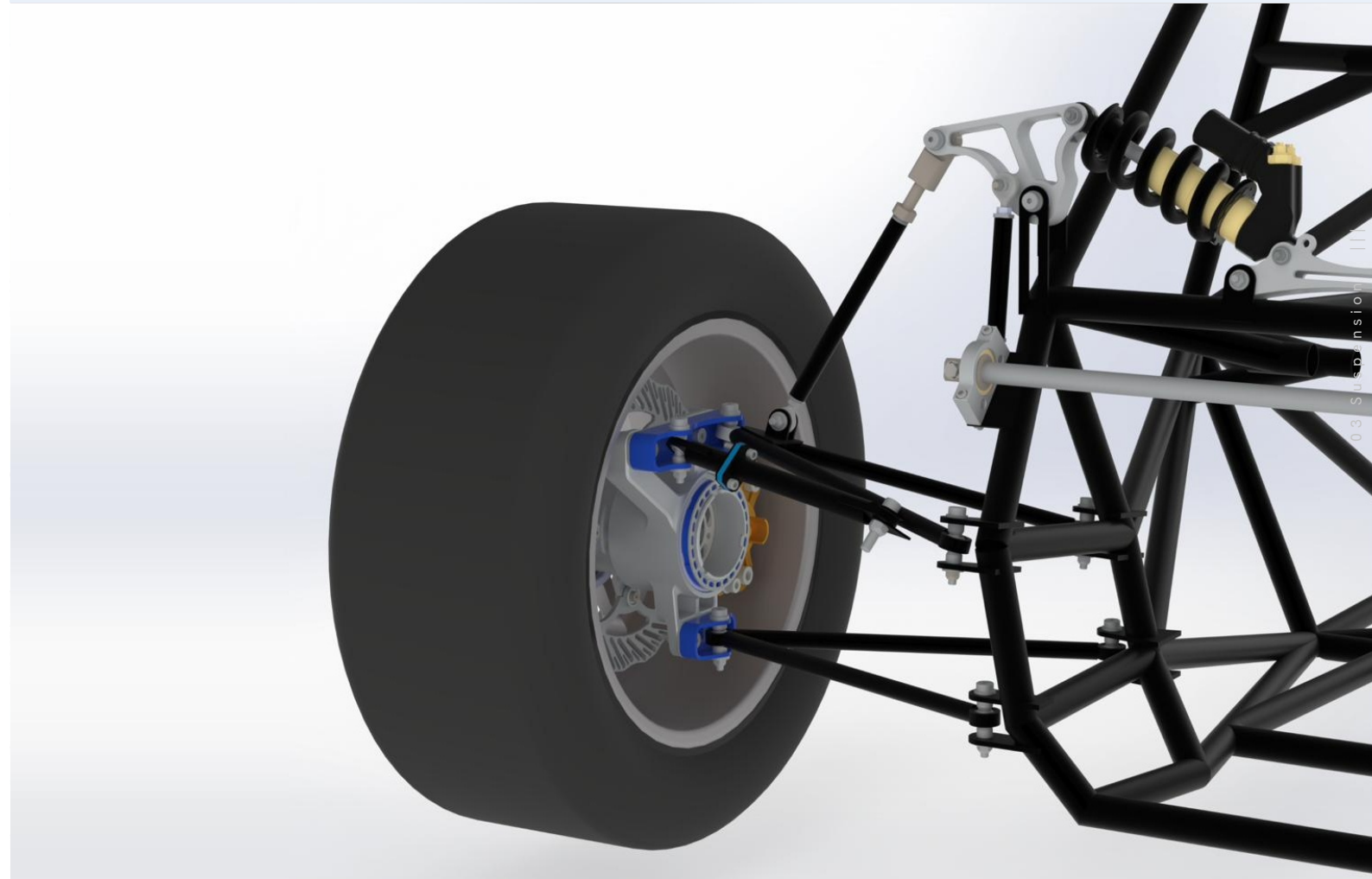
ROLE

Design owner, concept to track validation

TOOLS

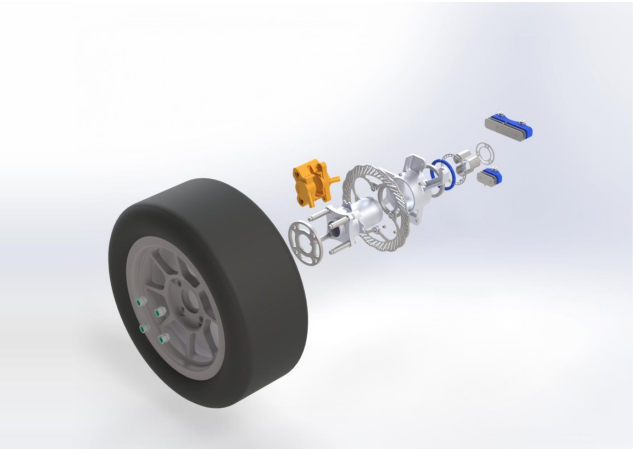
SolidWorks, Lotus Suspension Analysis, ANSYS Mechanical

SUSPENSION



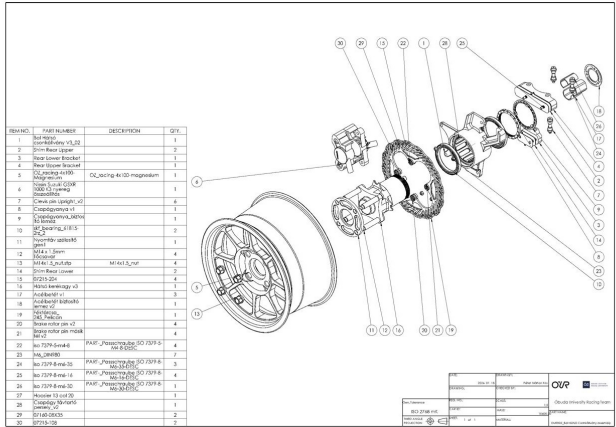
Rear corner assembly

Design decisions and validation



Exploded assembly

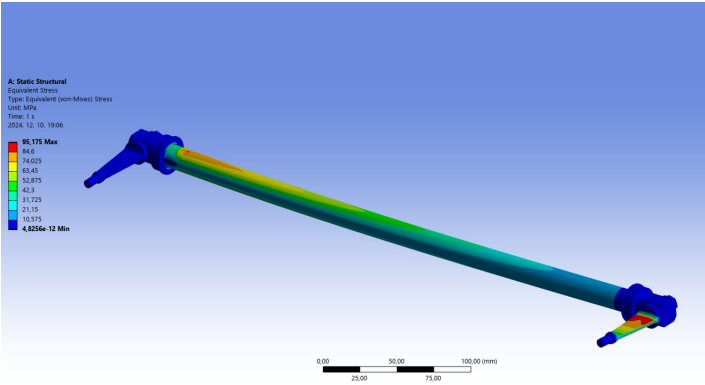
The machined aluminium upright packages its bearings in an O layout for better load support, with a lightweight spacer between them setting the pretension. Positive locking of the bearings against the wheel hub is solved with a designed aluminium bearing nut and retainer ring.



Assembly drawing, 30 item BOM

Thirty items split between machined parts and catalogue hardware: upright, hub, bearing nut, locking plate, floating disc bushings, brackets and shims against the 61815 bearing pair, the caliper, ISO 7379 shoulder screws and DIN 980 lock nuts.

SUSPENSION



Anti-roll bar blade, FEA

U-bar layout for tight packaging, designed to achieve a 0.5 deg/g roll gradient. Torsion tube on cylindrical supports so it can rotate, blade root fixed, 1100 N at the droplink. Sized to add 51.7 Nm/deg at the wheel and 894.7 Nm/deg at the bar on a 4.16 motion ratio; Ø16×14 tube gives 944 Nm/deg, blade 1366, peak stress 96 MPa.

Chassis

Óbuda University Racing Team · two generations
2024-2025

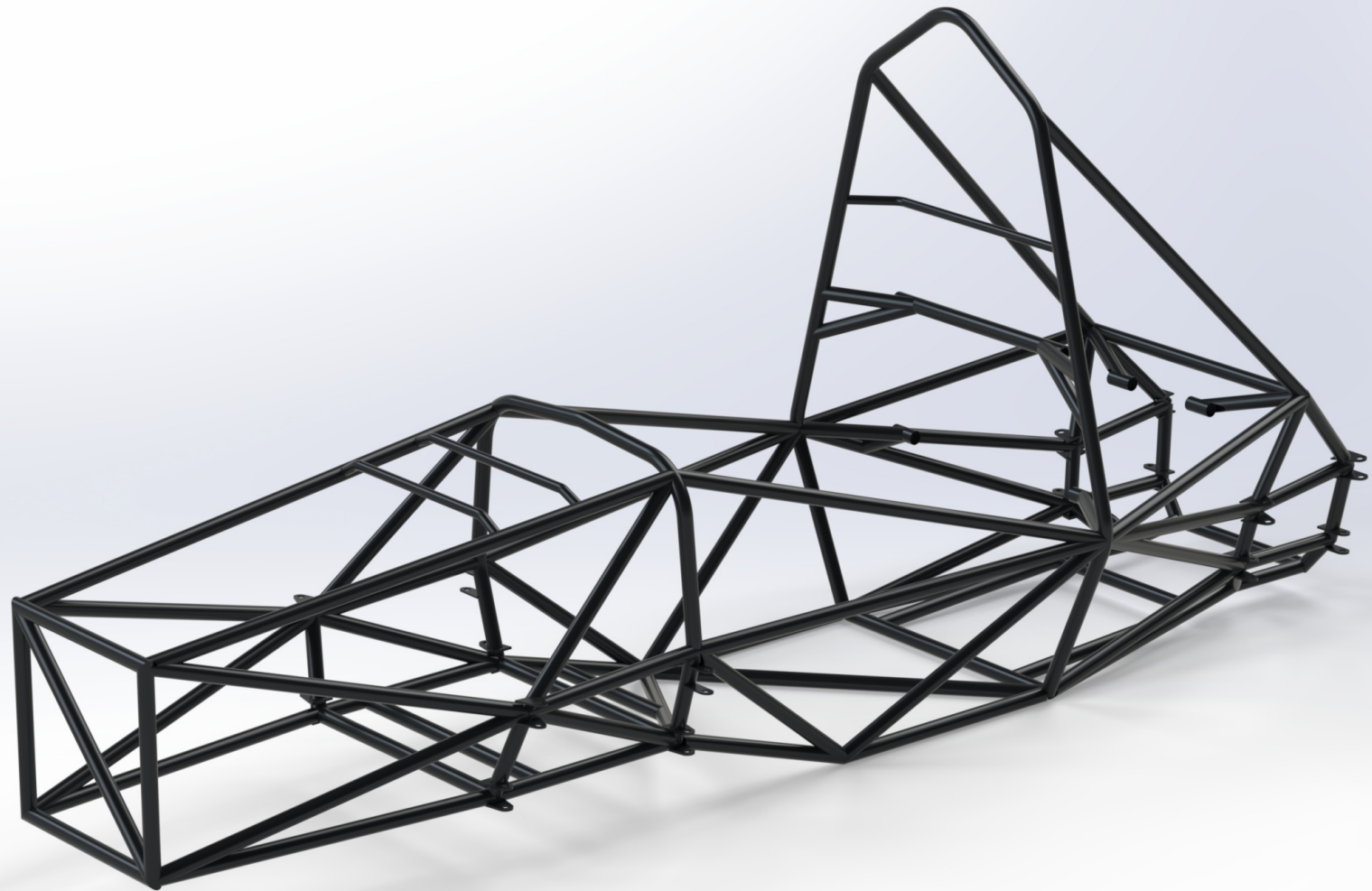
A rules-compliant steel spaceframe carrying suspension, damping system and powertrain loads at 4200 Nm/deg torsional stiffness, leaving suspension behaviour uncompromised while removing 10% of the mass compared to the previous chassis. Buildable on a jig the team could weld itself without losing pickup point accuracy.

ROLE

Frame design and subsystem packaging

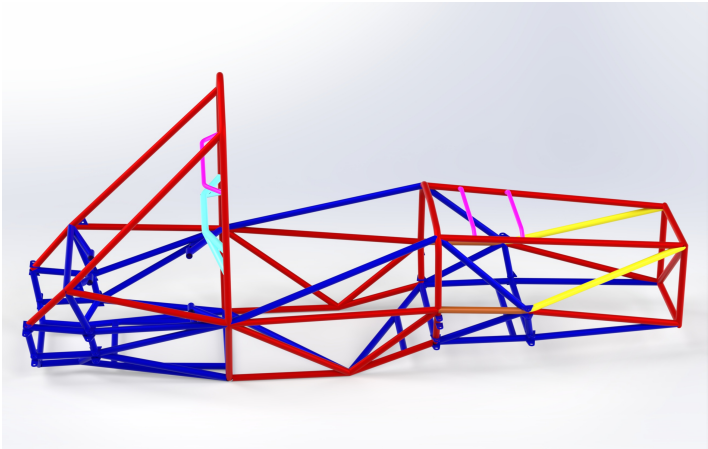
TOOLS

SolidWorks, ANSYS Mechanical



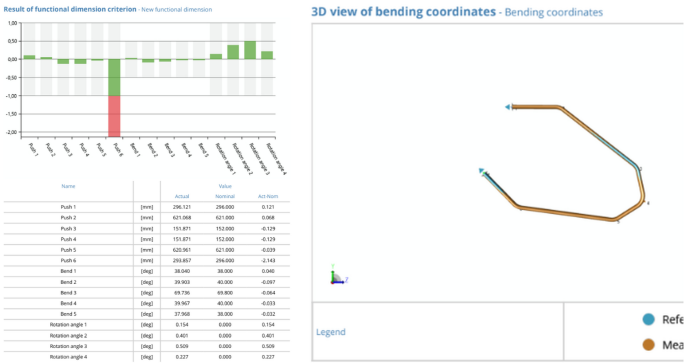
Tubular spaceframe chassis

Design decisions and validation



Tube layout by wall thickness

Frame modelled in SolidWorks Weldments with the node layout set by the suspension pickup points. E235 CR2 S2 throughout, hoops included, the colours grouping tubes by wall thickness and section as required to stay regulations compliant.



Bend trial, bender setup

One tube bent and measured against the CAD reference to learn how the section springs back, before any hoop was made. Feed lengths came out within 0.13 mm and bend angles within 0.10 deg; the deviations set the correction values used to bend the hoops.



Welded frame, as built

Tubes 2D laser cut and CNC bent with springback compensation, then TIG welded to process 141 on a Bosch profile jig that held the A-arm pickups and hoop feet in position through the whole weld sequence, so the frame came off the jig on geometry.

Powertrain

Óbuda University Racing Team · OUR003 · 2025

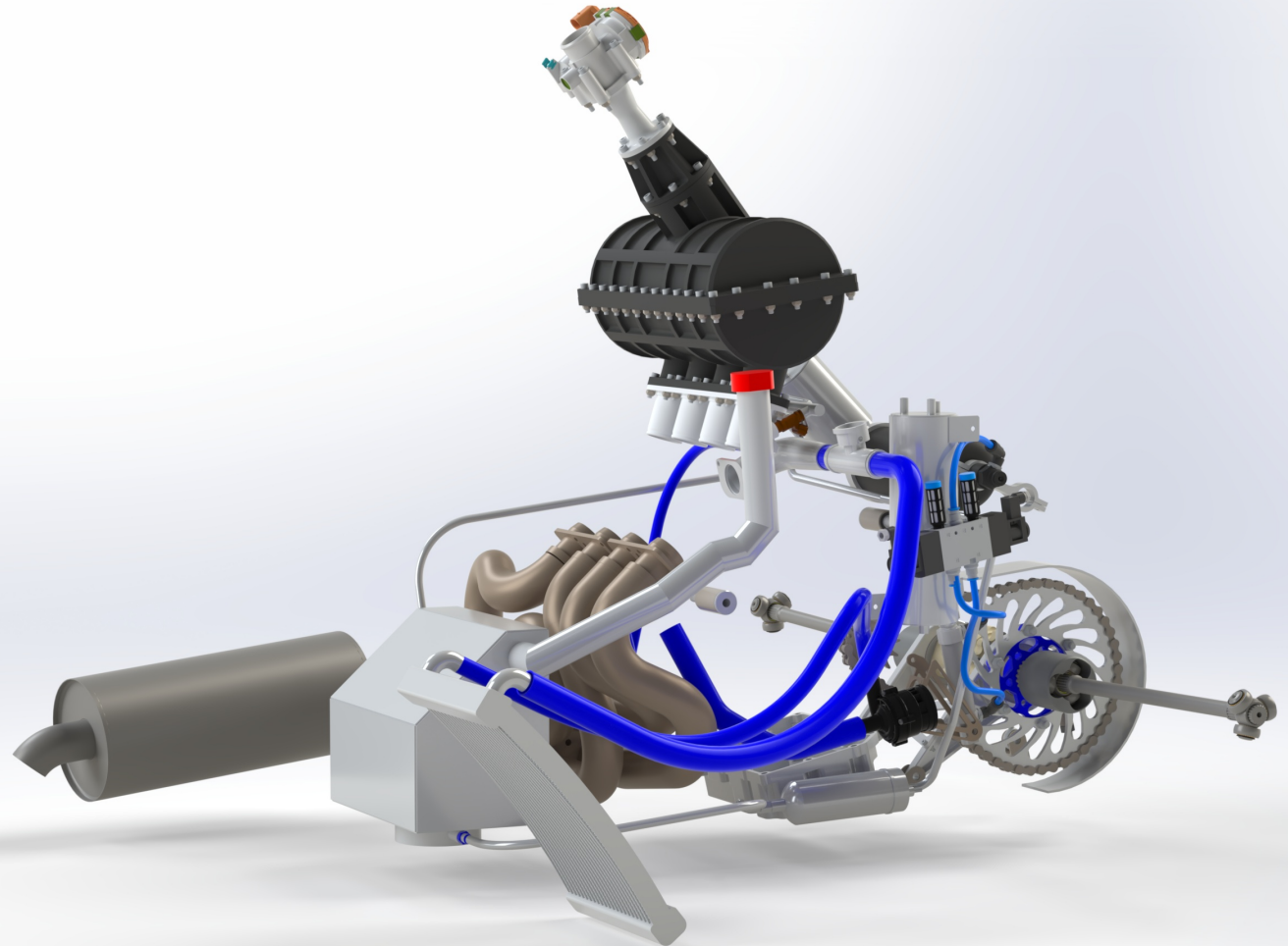
Suzuki GSX-R 600 K6 four cylinder engine based powertrain with pneumatic shifting mechanism for quick gear changes, behind the mandatory 20 mm intake restrictor. 4-1 manifold tuned for mid-range torque for better driveability across the dynamic disciplines of the competition, with the final drive taken through a chain to a limited slip differential. Tripod housings with halfshafts transferring the torque to the wheels. The powertrain concept achieved a 4.326 s acceleration result at Formula Student Alpe Adria 2025 and made the endurance finish possible at Formula Student Austria 2025.

ROLE

Designer and engine calibration

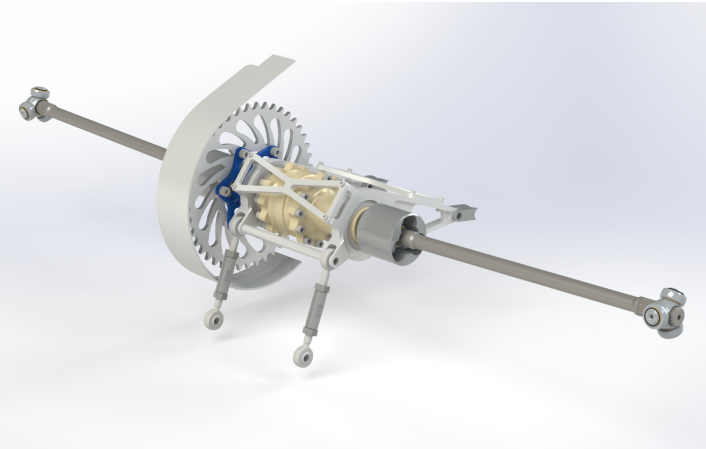
TOOLS

Lotus Engine Simulation, SolidWorks, engine dyno, MAXXECU MTune



Powertrain

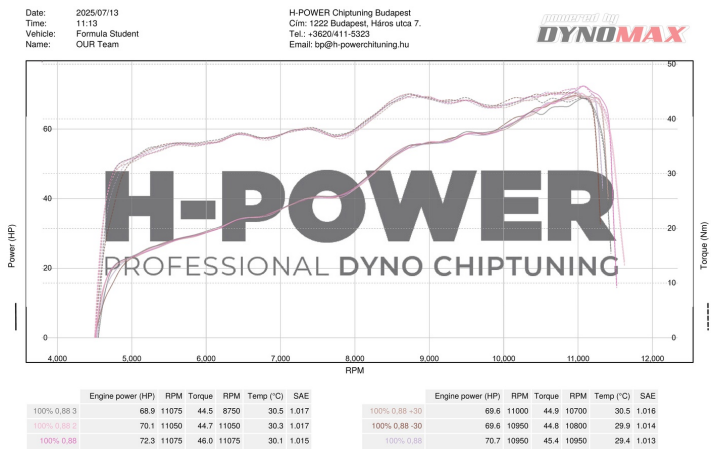
Design decisions and validation



Drivetrain

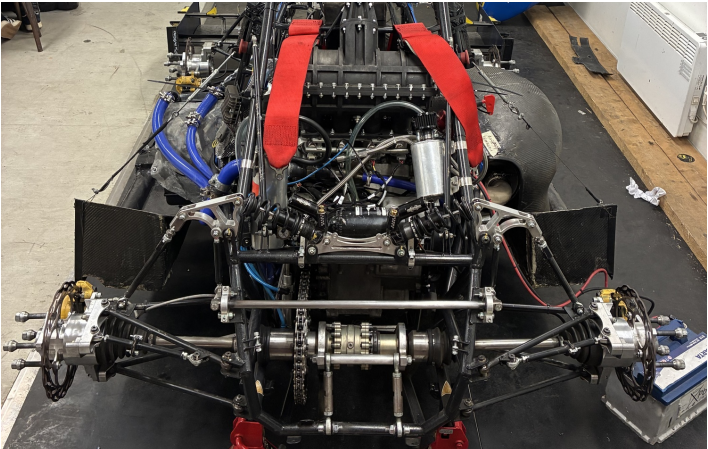
Weight reduced sprocket, optimised differential mounts and a 3.46 final drive ratio which, with the new power figures, gives an 18% performance gain in acceleration. Tripod joint inserts machined in house from 42CrMo4, designed for the higher torque figure and simpler assembly.

POWERTRAIN



Dyno result

Rolling dyno used to set ignition and fuel maps. Against the 20 mm restrictor the target was the flattest torque curve possible; the 6.2 l plenum works as an air buffer and cuts the drop at 7500-8000 rpm. Achieved 72 whp and 46 Nm at 11075 rpm, a 3 hp and 6 Nm gain on the previous design.



Powertrain packaging

GSX-R 600 K6 and final drive on polyurethane non-stressed mounts, isolating the engine from suspension loads. Titanium Grade 5 halfshafts reduced rotating mass by 19%, transferring torque through tripod joints into 42CrMo4 inserts in the wheelhubs.

Manufacturing

Óbuda University Racing Team · OUR002-OUR003
2023-2025

During the design phase every component geometry was defined to fit the manufacturing capabilities of the team’s in-house machines and its sponsors. The technical drawings for the rear upright, rear wheel hub, tripod housing, clevis pins and brake rotor pins were made to set up the NC code for the CNC lathe and mills, and carried enough information to machine the same parts on conventional machines if a sponsor could not finish a part in time. Machining in house gained two weeks over the competition, so the car was assembled and on track earlier, which returned 120 km more testing than in 2024.

ROLE
Design for manufacture, tooling and build support

MANUFACTURING TECHNOLOGIES	
3 axis CNC Mill	TIG 141
3 axis CNC Turning machine	2D CNC tube cutting
Conventional Mill and Turning machine	3D CNC tube bending

MANUFACTURING



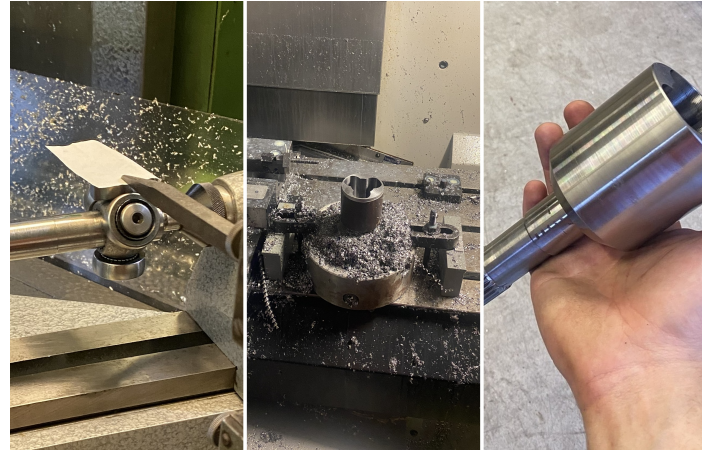
In-house machining

Processes owned in house



Wheel hub machining

Roughing of the round bar turned on a conventional lathe to take the weight out, then the bearing seat, the brake rotor pin flange, the radii and the bearing nut thread turned to diameter. The lug nut and rotor pin flange profiles and the tripod insert contour were then milled, and the bearing diameter finished last for the correct bearing fit.



Tripod housing machining

The three rollers of the chosen tripod joints were measured to define the contour inside the housing and the inserts in the hub, and the internal spline measured for the halfshaft spline design and for pairing a joint to each shaft, keeping axial play and imbalance to a minimum. The 42CrMo4 bar was roughed, the external spline cut, then the 50 mm deep internal contour milled in 0.1 mm steps to match the joints.



Clevis and brake rotor pin machining

Clevis and brake rotor pins are made in volume: 180 parts a year, run on the CNC lathe at 36 s a piece, so the full set is cut in two days once the design is frozen. The tooling for the turning process was designed according to the component design. The flexibility and quick machining time give us an edge on the assembly and make it possible to have spare parts.

Design index

Everything else, by subsystem

SUSPENSION

- Rear uprights
- Wheel hubs and bearing nuts
- Rockers, front and rear
- Pushrods and rod ends
- Anti-roll bar blades, torsion bar and mounts
- Damper mounts
- Clevis pins, spacers, shims

Thank you for going through the work. Every part shown was modelled, analysed, detailed to a drawing and run on the race car. If you would like to see the kinematic model behind it, feel free to contact me.

LINKS

Linkedin	linkedin.com/in/marton-peter-kiss
Render	drive.google.com/drive/folders/101EeCnLAeSsRCXe9H5hJj0ZTtAg6nLTU
Technical Drawings	github.com/kissmartonpeter/technical-drawings drive.google.com/drive/folders/1Bs_hG624g8UP7S-iSmn7peMOvbFwXAYE
Dyno	drive.google.com/drive/folders/1cVSiCJUvVFjgQMMcGQdymcF8vg5ppMd2

POWERTRAIN

- Restrictor and intake tube
- Upper and lower intake plenum
- Welded intake manifold
- 4-1 exhaust manifold and muffler
- Fuel tank assembly
- Sprocket and chainguard
- Differential mounts, tripod hubs, halfshafts

CHASSIS AND TOOLING

- Spaceframe, two generations
- Front and main hoop
- Suspension pickup brackets
- Bosch profile welding jig
- Assembly and setup table
- Full vehicle assembly drawings

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